Supplementary material SIRfyN

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1 Further variants and fuller comparisons

Fig. 1. *1 FID_∞ against MSD for various parameter choices for model selection.* **Left:** *PatchGAN adversary scale affects performance; the* **small** *receptive field is best.* **Center:** λ_B *encourages variations without much adverse affect on FID_∞; we chose $\lambda_B = 5$.* **Right:** $\lambda_P > 0$ *reliably produces more speculative models with weaker FID; the figure is Figure 4 from paper, for reference.*

Fig. 2. *Averaging for equivariance during inference hurts metrics. FID_∞ against MSD for* **red:** *256x256 images relighted using averaging at inference time;* **green:** *relighted 256x256 square crops;* **yellow:** *256x256 square crops from averaged images.*

Fig. 3. *No major effects result from using FID rather than FID_∞. FID against MSD for various parameter choices for model selection.* **Left:** *PatchGAN adversary scale affects performance; the* **small** *receptive field is best.* **Center:** λ_B *encourages variations without much adverse affect on FID_∞; we chose $\lambda_B = 5$.* **Right:** $\lambda_P > 0$ *reliably produces more speculative models with weaker FID. Compare Fig. 1.*

We describe here some natural variants that didn't make it into the main paper.

Equivariance: A function $\phi : \mathbf{x} \in X \to \mathbf{y} \in Y$ is equivariant under the action of a group G if there are actions of G on X and Y such that $\phi(g \circ \mathbf{x}) = g \circ \phi(\mathbf{x})$. Information being gained or lost at the boundary is an obstacle to applying the theory of group actions exactly (except for certain finite groups [**?**]). If one relaxes the definition to require only an approximate match, well-known visual feature representations tend to have strong equivariance properties either by design or in practice [**?**]. Imposing equivariance properties on intrinsic image decompositions (albedo at a point should not depend on image crop) shows sharp improvements [**?**]. Relighting has a form of equivariance property: small

movements of the camera frame should not affect the overall relighting of a scene (apart from rows and columns of the image moving in and out of view). We use that paper's crude averaging strategy because we know no better. In this approach we: pad an image; estimate multipliers for a number of closely overlapping crops of the padded image; average overlapping estimates to obtain final multipliers. Doing so reliably loses both MSD and FID_∞, for reasons we do not know (Figure 2).

Pixel uniformity loss: We expect there is some lighting that makes any location look bright (meaning the image value at each location should be bright), and that any location might be in darkness (meaning that the shading value at each location might be dark). Write $\uparrow M_V I_V$ for the largest value at each location over all generators of $M_V I_V$ and $\downarrow M_V S_V$ for the smallest value at each location over all generators of $M_V S_V$. We use:

$$\mathcal{L}_P(I_V; \phi) = \overline{E}$$

where

$$E = [-\min(0.95 - \uparrow M_V I_V, 0) + \max(\downarrow M_V S_V - 0.05, 0)]$$

Pixel uniformity loss reliably produces more speculative models with weaker FID_∞ (Figure 1).

Fig. 4. *Images relighted at $\alpha = 0.1$ (more variable;* **left***; cf $\alpha = 0.5$, less variable, $\alpha = 0.01$, much more variable, in Fig 5) and the first five multiplier fields for each* **right***. Notice how the multipliers are keyed to structures in the image – the method "knows" some geometry and radiometry. Reproduced from Fig 3 of main paper, without arrows, for reference.*

Fig. 5. *Images relighted at $\alpha = 0.5$* **left***, less variable and $\alpha = 0.01$, much more variable* **right***. At $\alpha = 0.5$ images have higher quality, but tend to be less variable. At $\alpha = 0.01$, the difficulties presented by extrapolation become more obvious.*

Fig. 6. *Our method produces qualitative changes in the distribution of light, and appears to generalize well across datasets.* **Left:** *Further qualitative results on LSUN-Bedrooms;* **Right:** *Results for the first image each of the validation set for 10 scenes with "room" in their name from the PlacesRooms dataset; relights are* not *curated.* **Top:** *Image;* **other:** *relights obtained using our method, $\alpha = 0.5$. All relights are shown at the same mean intensity as the original image, to ensure that effects are not masked by overall shift in brightness.*

Fig. 7. *We show all pairs of source/target/least-squares images used to prepare Figure 5 of the main paper. Recall that shows results of: rendering CGI scenes under two lights; computing the multipliers for the* **source***; constructing weighted sum of multipliers that yields a least squares fit to the* **target***; then comparing target to the* **least squares***. The* **residual** *error in intensity is shown on a scale where white is 1; mid grey is 0; and black is -1. Note relatively good fits, suggesting that for the cases we have investigated, multipliers are a reasonable fit.* **Quantitative:** *over a total of 11 pairs (22 tests), for a dataset we will provide, we find average metrics between the least squares prediction and the target of RMSD: 0.0503 PSNR: 26.6 LPIPS: 0.0648, suggesting quite successful prediction of the target by source multipliers.*

2 Theorem 1: Similar scenes have similar radiosity

We wish to compare the radiosity of a scene with that of a similar scene, of somewhat different geometry, albedo and luminaire. The key result is the difference between these two radiosities can be described in a sensible way, and can be bounded. In turn, it follows that sufficiently similar scenes will share generators.

2.1 Notation

It is enough to work with a simplified model of a scene V as a tuple $(\mathcal{G}, \rho, E)$ of geometry, albedo and luminaire (rather than stochastic family of luminaires). $\mathcal{G}$ consists of a set of surfaces parametrized in some way and a 2D domain for that parametrization; write $\mathcal{G} = (\mathbf{s}(\mathbf{x}), \mathcal{D})$. Significant complications in the analysis can occur associated with concave "creases" or at surface interpenetrations. These complications aren't physically important – every corner in a room is made smooth by at least a thin layer of paint or dirt – and so we avoid them by assuming:

- $\mathcal{D}$ is compact (but need not be connected) with finite area.
- Each connected component of $\mathcal{D}$ represents a distinct surface.
- Surfaces are parametrized by $\mathbf{s}(\mathbf{x})$, where $\mathbf{s}$ is at least C^1 and is an embedding (surfaces do not self-intersect).
- Surfaces are orientable, and are oriented.
- For any pair of points $\mathbf{x}, \mathbf{y}$, we have $k(\mathbf{x}, \mathbf{y}) \leq C$ for some large finite constant C and k defined below. This rules out "throats" – places where the gap between surfaces narrows and then opens out, Figure 8 – in the geometry, and bounds the largest curvatures. Larger C corresponds to narrower throats and tighter curvatures being allowed. The condition is sufficient to ensure the kernel is compact, below.

Note that parametrizing surfaces by atlases of charts complicates notation without changing the analysis; we will suppress these details. Write

$$\| E \|_p = \left[\int_{\mathcal{D}} E^p d\mathbf{x} \right]^{1/p}$$

$(1 \leq p \leq \infty)$ for the p norm of E. Recall that for an operator $\mathbf{K}$

$$\| \mathbf{K} \|_p \leq \| \mathbf{K} \|_\infty^{\frac{(p-1)}{p}} \| \mathbf{K} \|_1^{\frac{1}{p}}$$

(eg [?], [?] p.144 or [?] p.518). All operators here will have both $\| \mathbf{K} \|_1 < 1$ and $\| \mathbf{K} \|_\infty < 1$, so that results apply for any p; we write $\| \mathbf{K} \|$.

We wish to compare the radiosity B_V of a scene V with the radiosity B'_V of a similar scene V'. These scenes have somewhat different geometry, albedo and luminaire. We consider affine transformations of the geometry, so that $\mathcal{G}'(\mathcal{G}, \mathcal{A}, \mathbf{b}) = (\mathcal{A}\mathbf{s}(\mathbf{x}) + \mathbf{b}, \mathcal{D})$. We will prove:

Theorem 1: *For $V = (\mathcal{G}, \rho, E)$ and $V' = (\mathcal{G}'(\mathcal{G}, \mathcal{A}, \mathbf{b}), \rho', E')$, where $\epsilon_E \| E \| = \| E - E' \|$, $\epsilon_\rho = sup_{\mathcal{D}} abs[\rho - \rho']$, $p = sup_{\mathcal{D}} \rho$, $p' = sup_{\mathcal{D}} \rho'$, c is the condition number of $\mathcal{A}$ (ratio of largest to smallest eigenvalues), we have:*

$$\| B_V - B_{V'} \| \leq \left[\frac{\epsilon_E}{1-p} + \left[\frac{\epsilon_\rho (1+\epsilon_E)}{(1-p)(1-p')} \right] + \frac{(c^4-1)(1+\epsilon_E)}{(1-p')^2} \right] \| E \|$$

2.2 Standard results

Fig. 8. Left: *Our notation.* **Right:** *a throat.*

The geometry consists of a set of parametrized surfaces. Write $\mathcal{D}$ for the domain of the parametrization. For a parameter vector $\mathbf{x} \in \mathcal{D}$ write $\mathbf{s}(\mathbf{x})$ for the corresponding surface point; $B(\mathbf{x})$ for radiosity at that point; $\rho(\mathbf{x})$ for the albedo at that point; $E(\mathbf{x})$ for the radiant exitance at that point; $V(\mathbf{x}, \mathbf{y})$ for the *visibility function* that is 1 when there is a direct line of sight from $\mathbf{s}(\mathbf{x})$ to $\mathbf{s}(\mathbf{x})$ and 0 otherwise; $\mathbf{d}_{\mathbf{x}\to\mathbf{y}} = \mathbf{s}(\mathbf{y}) - \mathbf{s}(\mathbf{x})$; and $\mathbf{N}(\mathbf{x})$ for a unit normal at $\mathbf{x}$. Figure 8 shows some notation commonly used. Standard diffuse interreflection theory (see, eg [?,?,?,?]) gives

$$B(\mathbf{x}) = E(\mathbf{x}) + \rho(\mathbf{x}) \int_{\mathcal{D}} k(\mathbf{x}, \mathbf{y}) B(\mathbf{y}) dA_{\mathbf{y}}$$

where

$$\begin{aligned} k(\mathbf{x}, \mathbf{y}) &= \frac{1}{\pi} \frac{V(\mathbf{x}, \mathbf{y}) \cos\theta_{\mathbf{x}} \cos\theta_{\mathbf{y}}}{\| \mathbf{d}_{\mathbf{x}\to\mathbf{y}} \|_2^2} \\ &\qquad \text{(notation in Figure 8)} \\ &= \frac{V(\mathbf{x}, \mathbf{y}) \left[\mathbf{N}^T(\mathbf{x}) \mathbf{d}_{\mathbf{x}\to\mathbf{y}} \right] \left[\mathbf{N}^T(\mathbf{y}) \mathbf{d}_{\mathbf{y}\to\mathbf{x}} \right]}{\left[\mathbf{d}_{\mathbf{x}\to\mathbf{y}}^T \mathbf{d}_{\mathbf{x}\to\mathbf{y}} \right]^2} \end{aligned}$$

(recalling $\mathbf{N}$ is a unit normal). Write $\mathbf{D}_\rho$ for the linear operator that maps $f(\mathbf{x})$ to $(\rho f)(\mathbf{x})$. We can now abbreviate into $B = E + \mathbf{D}_\rho \mathbf{T} B = E + \mathbf{K} B$, where $\mathbf{D}_\rho$, $\mathbf{T}$ and $\mathbf{K}$ are linear integral operators. It is known that

$$\| \mathbf{T} \| \leq 1$$

(for example, our $\mathbf{T}$ is the $\mathbf{KG}$ of [?], sec 5.3, with constant diffuse BRDF of 1). Furthermore, writing

$$p = \| \rho(\mathbf{x}) \|_\infty < 1$$

we have

$$\| \mathbf{D}_\rho \| \leq p$$

by simple calculation and so

$$\|\mathbf{K}\| \leq p.$$

For our assumptions,

$$\int_{\mathcal{D}}\int_{\mathcal{D}} |k(\mathbf{x},\mathbf{y})|^2 \, dA_{\mathbf{x}} dA_{\mathbf{y}} \leq C^2 \text{Area}(\mathcal{D})^2,$$

so that k is a Hilbert-Schmidt kernel. Because the kernel is Hilbert-Schmidt, $\mathbf{K}$ is a compact operator (eg [**?**]). There is then a Neumann series solution

$$\begin{aligned} B &= \mathbf{M}^{-1}E \\ &= E + \mathbf{K}E + \mathbf{K}^2E + \ldots \\ &= \text{exitance} + \text{one bounce} + \text{two bounces} + \ldots \end{aligned}$$

which converges for all $r < 1$ (and so all practical scenes). It follows that

$$\|\mathbf{M}^{-1}\| \leq \frac{1}{1-p}$$

2.3 Lemma 1: Perturbed Luminaire

Assume that the luminaires for V' differ from those of V (but geometry and albedo are the same), so $V = (\mathcal{G}, \rho, E)$ and $V' = (\mathcal{G}, \rho, E')$. Standard results yield:

Lemma 1: *using the notation above*

$$\|B_V - B_{V'}\| \leq \frac{1}{1-p}\|E - E'\| = \frac{\epsilon_E}{1-p}\|E\|.$$

Proof: $\|\mathbf{M}^{-1}\|$ is bounded by $\frac{1}{1-p}$, above. □

2.4 Lemma 3: Perturbed Albedo

Now assume that $V = (\mathcal{G}, \rho, E)$ and $V' = (\mathcal{G}, \rho', E)$ and $\rho' = \rho + \delta$. Write $\epsilon_\rho = \|\mathbf{D}_\delta\|$. We expect B' to differ from B_V because it contains multiple bounce terms (above). For example, the shading in a very dark room is notably different from that in a very light room (eg [**?**,**?**,**?**]), because low albedos mean that the multiple bounce terms are very small. Equivalently $\mathbf{M}$ and $\mathbf{M}'$ are different. However, $\mathbf{G}$ is bounded and we have $\|\mathbf{M}\| = p \leq 1$ and $\|\mathbf{M}'\| = p' \leq 1$. The dynamic range of practical albedos is quite limited (eg [**?**]), so it is reasonable to bound $\rho(\mathbf{x})$ away from both 0 and 1.

Lemma 2: *using the notation above, for all pairs of* $0 < \epsilon \leq \rho \leq (1-\epsilon) < 1$ *and* $0 < \epsilon \leq \rho' \leq (1-\epsilon) < 1$, $\|\mathbf{D}_{\rho-\rho'}\| = \epsilon_\rho < 1$

Proof: for some test function f,

$$\mathbf{D}_{\rho-\rho'} f = (\rho(\mathbf{x}) - \rho'(\mathbf{x}))f(\mathbf{x}).$$

Then

$$\begin{aligned}\|\mathbf{D}_{\rho-\rho'} f\|_1 &= \int_{\mathcal{D}} |(\rho(\mathbf{x}) - \rho'(\mathbf{x}))f(\mathbf{x})|\, dA_{\mathbf{x}} \\ &\leq \int_{\mathcal{D}} |(\rho(\mathbf{x}) - \rho'(\mathbf{x}))||f(\mathbf{x})|\, dA_{\mathbf{x}} \\ &\leq \sup_{\mathbf{x}} [|(\rho(\mathbf{x}) - \rho'(\mathbf{x}))|] \int |f(\mathbf{x})|\, dA_{\mathbf{x}}.\end{aligned}$$

The constraints on ρ, ρ' ensure

$$\sup \mathbf{x} [|(\rho(\mathbf{x}) - \rho'(\mathbf{x}))|] < 1$$

and $\|\mathbf{D}_{\rho-\rho'}\|_\infty < 1$, so the result follows for all norms. □

This yields the novel (but fairly straightforward)

Lemma 3: *using the notation above*

$$\|B_{V'} - B_V\| \leq \left[\frac{1}{1-p}\right]\left[\frac{1}{1-p'}\right]\epsilon_\rho \|E\|$$

Proof: We have $(I - \mathbf{D}_\rho \mathbf{G})B_V = E = (I - \mathbf{D}_{(\rho+\delta)}\mathbf{G})B_{V'}$, so

$$\begin{aligned}B_{V'} - B_V &= \mathbf{D}_{\rho'}\mathbf{G}B_{V'} - \mathbf{D}_\rho \mathbf{G}B_V \\ &= \mathbf{D}_{\rho'}\mathbf{G}B_V - \mathbf{D}_\rho \mathbf{G}B_V + \mathbf{D}_{\rho'}\mathbf{G}(B_{V'} - B_V) \\ &= \mathbf{D}_\delta \mathbf{G}B_V + \mathbf{D}_{\rho'}\mathbf{G}(B_{V'} - B_V) \\ &= \mathbf{D}_\delta \mathbf{G}\mathbf{M}^{-1}E + \mathbf{D}_{\rho'}\mathbf{G}(B_{V'} - B_V)\end{aligned}$$

so

$$B_{V'} - B_V = \mathbf{M}'^{-1}\mathbf{D}_\delta \mathbf{G}\mathbf{M}^{-1}E.$$

Now $\|\mathbf{D}_\delta \mathbf{G}\| \leq \epsilon_\rho$, so

$$\begin{aligned}\|B_{V'} - B_V\| &= \|\mathbf{M}'^{-1}\mathbf{D}_\delta \mathbf{G}\mathbf{M}^{-1}E\| \\ &\leq \left[\frac{1}{1-p}\right]\left[\frac{1}{1-p'}\right]\epsilon_\rho \|E\|\end{aligned}$$

□

2.5 Theorem 1: Perturbed Geometry

We wish to study the effect of small changes in geometry on the solution B. We have found no results on the effect of perturbing geometry, which appear to be hard to obtain (cf [**?**], p176 says they are; [**?**] estimate the effect of flat ports on an integrating sphere).

Small geometric changes can produce major changes in visibility which are difficult to bound. We consider affine transformations of the surfaces, so that $\mathbf{s}(\mathbf{x})$ is mapped to $\mathcal{T}(\mathbf{s}) = \mathcal{A}\mathbf{s}(\mathbf{x}) + \mathbf{b}$ ($\det(\mathcal{A}) \geq 0$). We assume that the smallest singular value of $\mathcal{A}$ is positive. There are two advantages of this model: first, it is a reasonable description of small changes between otherwise quite similar rooms (small anisotropic scalings in the right coordinate system); second, the visibility function for $\mathcal{T}(\mathbf{s})$ is the same as the visibility function for $\mathbf{s}$ (lemma 4, below).

Transformations and Functions The transformation produces a natural bijection between functions of original and transformed geometry. This is so natural that it can be difficult to see at first glance. Assume we have a function f defined on the first geometry. We must know the value that f takes at each point on the geometry, which means we know the value at each point in the parameter space. But this parameter space parametrizes the transformed geometry, too, so we have a corresponding function on the transformed geometry (this argument yields that the mapping between function spaces is a bijection). For f a function defined on the original parametrization, write $\mathcal{T}(f)$ for the transformed version.

This natural correspondence may not respect the geometric transformation on the embedded surfaces. This means we need to distinguish between a function mapped through the correspondence and one evaluated on the new geometry. For example, consider the function $\text{sqdist}(\mathbf{x}_0, \mathbf{x}_1) = \|\mathbf{s}(\mathbf{x}_0) - \mathbf{s}(\mathbf{x}_1)\|_2^2$. If $\mathcal{A} = 2 * Id$, the distance between points on the transformed geometry has doubled. Write $\text{sqdist}'(\mathbf{x}_0, \mathbf{x}_1) = \|\mathcal{A}\mathbf{s}(\mathbf{x}_0) - \mathcal{A}\mathbf{s}(\mathbf{x}_1)\|_2^2$ (the prime indicates the function is evaluated on the transformed geometry). We have $\mathcal{T}(\text{sqdist})(\mathbf{x}_0, \mathbf{x}_1) = (1/4)\|\mathcal{A}\mathbf{s}(\mathbf{x}_0) - \mathcal{A}\mathbf{s}(\mathbf{x}_1)\|_2^2 = (1/4)\text{sqdist}'(\mathbf{x}_0, \mathbf{x}_1)$.

Transformations and the Kernel A similar natural correspondence applies to operators. We must now consider the relationship between $\mathcal{T}(\mathbf{K})$ and $\mathbf{K}'$. Each is a bounded linear operator defined on functions of $\mathcal{G}'$, but they are different. $\mathcal{T}(\mathbf{K})$ is obtained by applying the natural correspondence to $\mathcal{K}$; in contrast $\mathcal{K}'$ is obtained by computing the kernel *using the geometry of* $\mathcal{G}'$, which may give a different operation.

By tracking the effects of affine transformations on the terms of k, we can constrain the difference. First, we have

Lemma 4: $V(\mathbf{x}, \mathbf{y}) = \mathcal{T}(V(\mathbf{x}, \mathbf{y})) = V'(\mathbf{x}, \mathbf{y})$

Proof: The main issue is keeping track of notation; the first equality is obvious (natural correspondence); the second – that computing the view function in the transformed space is equivalent to computing it in the original space, and mapping through correspondence – follows because affine transformations map lines to lines and preserve incidence. □

We must now investigate how k transforms. We have:

Lemma 5: $\mathbf{N}'(\mathbf{x}) = \frac{\mathcal{A}^{-T}\mathbf{N_x}}{\sqrt{\mathbf{N}^T\mathcal{A}^{-1}\mathcal{A}^{-T}\mathbf{N}}}$

Proof: Write $\Pi_{\mathbf{x}}$ for the tangent plane at $\mathbf{x}$. For $\mathbf{u} \in \Pi_{\mathbf{x}}$, we have for non-unit normal $\mathbf{n}$, $\mathbf{n}(\mathbf{x})^T(\mathbf{u}-\mathbf{x}) = 0$. But $\mathbf{u}$ transforms to $\mathcal{A}\mathbf{u}+\mathbf{b}$, etc. and we must have $\mathbf{n}'(\mathbf{x})^T(\mathcal{A}(\mathbf{u}-\mathbf{x})) = 0$, and the result follows by requiring the normal be a unit normal; equivalently, normals are covariant. □

Write σ_x for the largest singular value of $\mathcal{A}$, σ_n for the smallest (which is positive), and $c = \frac{\sigma_x}{\sigma_n}$ for the condition number. We have

Lemma 6: $(\sigma_n)\mathbf{N}^T\mathbf{d}_{\mathbf{u}\to\mathbf{v}} \leq \mathbf{N}'^T\mathbf{d}'_{\mathbf{u}\to\mathbf{v}} \leq (\sigma_x)\mathbf{N}^T\mathbf{d}_{\mathbf{u}\to\mathbf{v}}$

Proof: $\mathbf{N}'^T\mathbf{d}'_{\mathbf{u}\to\mathbf{v}} = \frac{\mathbf{N}^T\mathbf{A}^{-1}\mathbf{A}\mathbf{d}_{\mathbf{u}\to\mathbf{v}}}{\sqrt{\mathbf{N}^T\mathbf{A}^{-1}\mathbf{A}^{-T}\mathbf{N}}}$ and the bound follows. □

Lemma 7: ${\sigma_n}^2\mathcal{T}(dA_{\mathbf{y}}) \leq dA'_{\mathbf{y}} \leq {\sigma_x}^2\mathcal{T}(dA_{\mathbf{y}})$.

Proof: $dA'_{\mathbf{y}}$ is the infinitesimal area spanned by two infinitesimal vectors dx_1 and dx_2. The largest (resp. smallest) scaling of each length is by the largest (resp. smallest) singular value of $\mathcal{A}$. □

We then obtain the novel:

Lemma 8: *For affine transformations,* $c^{-4}\mathbf{K}(\mathcal{T}(f)) \leq \mathbf{K}'(\mathcal{T}(f)) \leq c^4\mathbf{K}(\mathcal{T}(f))$ *for any* $f_{\mathbf{s}}$

Proof: For any two vectors $\mathbf{u}'^T\mathbf{v}' = \mathbf{u}^T\mathcal{A}^T\mathcal{A}\mathbf{v}$. Then $\sigma_n^2\mathbf{u}^T\mathbf{v} \leq \mathbf{u}^T\mathcal{A}^T\mathcal{A}\mathbf{v} = \mathbf{u}'^T\mathbf{v}' \leq \sigma_x^2\mathbf{u}^T\mathbf{v}$. Lemma 6 deals with the angles. Write

$$\mathbf{K}'(\mathcal{T}(f)) = \int_{\mathcal{D}} k'(\mathbf{x},\mathbf{y})f(\mathbf{y})dA'_{\mathbf{y}}$$

where

$$k'(\mathbf{x},\mathbf{y}) = \frac{V'(\mathbf{x},\mathbf{y})\left[\mathbf{N}'^T(\mathbf{x})\mathbf{d}'_{\mathbf{x}\to\mathbf{y}}\right]\left[\mathbf{N}'^T(\mathbf{y})\mathbf{d}'_{\mathbf{y}\to\mathbf{x}}\right]}{\left[\mathbf{d}'^T_{\mathbf{x}\to\mathbf{y}}\mathbf{d}'_{\mathbf{x}\to\mathbf{y}}\right]^2}$$

But

$$\begin{aligned}\begin{bmatrix}\left[(\sigma_n)\mathbf{N}^T(\mathbf{x})\mathbf{d}_{\mathbf{x}\to\mathbf{y}}\right]\times\\ \left[(\sigma_n)\mathbf{N}^T(\mathbf{y})\mathbf{d}_{\mathbf{y}\to\mathbf{x}}\right]\times\\ \left[\sigma_n^2 dA_{\mathbf{y}}\right]\end{bmatrix} &\le \begin{bmatrix}\left[\mathbf{N}'^T(\mathbf{x})\mathbf{d}'_{\mathbf{x}\to\mathbf{y}}\right]\times\\ \left[\mathbf{N}'^T(\mathbf{x})\mathbf{d}'_{\mathbf{y}\to\mathbf{x}}\right]\times\\ dA'_{\mathbf{y}}\end{bmatrix}\\ &\le \begin{bmatrix}\left[(\sigma_x)\mathbf{N}^T(\mathbf{x})\mathbf{d}_{\mathbf{x}\to\mathbf{y}}\right]\times\\ \left[(\sigma_x)\mathbf{N}^T(\mathbf{y})\mathbf{d}_{\mathbf{y}\to\mathbf{x}}\right]\times\\ \left[\sigma_x^2 dA_{\mathbf{y}}\right]\end{bmatrix}\end{aligned}$$

and

$$\begin{aligned}\sigma_n^4\left[\mathbf{d}^T_{\mathbf{x}\to\mathbf{y}}\mathbf{d}_{\mathbf{x}\to\mathbf{y}}\right]^2 &\le \left[\mathbf{d}'^T_{\mathbf{x}\to\mathbf{y}}\mathbf{d}'_{\mathbf{x}\to\mathbf{y}}\right]^2\\ &= \left[\mathbf{d}^T_{\mathbf{x}\to\mathbf{y}}\mathcal{A}^T\mathcal{A}\mathbf{d}_{\mathbf{x}\to\mathbf{y}}\right]^2\\ &\le \sigma_x^4\left[\mathbf{d}^T_{\mathbf{x}\to\mathbf{y}}\mathbf{d}_{\mathbf{x}\to\mathbf{y}}\right]^2\end{aligned}$$

so that

$$\begin{aligned}c^{-4}\mathcal{T}(\mathbf{K})(\mathcal{T}(f)) &= \int_{\mathcal{D}}\frac{\sigma_n^4}{\sigma_x^4}k(\mathbf{x},\mathbf{y})f(\mathbf{y})dA_{\mathbf{y}}\\ &\le \int_{\mathcal{D}}k'(\mathbf{x},\mathbf{y})f(\mathbf{y})dA'_{\mathbf{y}}\\ &\le \int_{\mathcal{D}}\frac{\sigma_x^4}{\sigma_n^4}k(\mathbf{x},\mathbf{y})f(\mathbf{y})dA_{\mathbf{y}}\\ &= c^4\mathcal{T}(\mathbf{K})(\mathcal{T}(f))\end{aligned}$$

□.

which leads to

Lemma 9: *With the notation above,*

$$\|B_V - B_{V'}\| \le \frac{(c^4-1)}{(1-p)^2}\|E\|$$

Proof:

$$\begin{aligned}B_{V'} - B_V &= (\mathcal{T}(E) - E) + (\mathbf{K}'B_{V'} - \mathbf{K}B_V)\\ &= \mathbf{K}'B_{V'} - \mathbf{K}'B_V + \mathbf{K}'B_V - \mathbf{K}B_V\\ &\quad (\text{recall } \mathcal{T}(E) = E)\\ &= \mathbf{K}'(B_{V'} - B_V) + \mathbf{K}'B_V - \mathbf{K}B_V\end{aligned}$$

so that

$$(\mathbf{I} - \mathbf{K}')\,(B_{V'} - B_V) = (\mathbf{K}' - \mathbf{K})\,B_V$$

and

$$(B_{V'} - B_V) = \mathbf{M}'^{-1}\,(\mathbf{K}' - \mathbf{K})\,B_V.$$

Notice that the kernel is everywhere non-negative, and for every point $\mathbf{x}$ and for any non-negative function f, we have $0 \leq (\mathbf{K}f)(x)$. Since E is non-negative, so is B_V. From lemma 6, for non-negative f, we have $(\mathbf{K}'f)(\mathbf{x}) \leq c^4(\mathbf{K}f)(\mathbf{x})$ at every point $\mathbf{x}$, so that $\mathbf{K}'f - \mathbf{K}f \leq (c^4 - 1)\mathbf{K}f$ at every point. This means

$$\begin{aligned}
\|\,(B_{V'} - B_V)\,\| &= \|\,\mathbf{M}'^{-1}\,(\mathbf{K}' - \mathbf{K})\,B_V\,\| \\
&\leq \frac{1}{(1-p')}\|\,(\mathbf{K}' - \mathbf{K})\,B_V\,\| \\
&\leq \frac{1}{(1-p')}(c^4 - 1)\|\,B_V\,\| \\
&\leq \frac{1}{(1-p')}(c^4 - 1)\frac{1}{(1-p)}\|\,E\,\| \\
&= \frac{(c^4 - 1)}{(1-p)^2}\|\,E\,\|
\end{aligned}$$

where the last step works because $p' = p$, because $\rho' = \mathcal{T}(\rho) = \rho$. □.

Finally, we can prove the entirely novel

Theorem 1: *For $V = (\mathcal{G}, \rho, E)$ and $V' = (\mathcal{G}'(\mathcal{G}, \mathcal{A}, \mathbf{b}), \rho', E')$, where $\epsilon_E\|\,E\,\| = \|\,E - E'\,\|$, $\epsilon_\rho = sup_{\mathcal{D}}abs[\rho - \rho']$, $p = sup_{\mathcal{D}}\rho$, $p' = sup_{\mathcal{D}}\rho'$, c is the condition number of $\mathcal{A}$ (ratio of largest to smallest eigenvalues), we have:*

$$\|\,B_V - B_{V'}\,\| \leq \left[\left[\frac{\epsilon_E}{1-p}\right] + \left[\frac{\epsilon_\rho(1+\epsilon_E)}{(1-p)(1-p')}\right] + \frac{(c^4-1)(1+\epsilon_E)}{(1-p')^2}\right]\|\,E\,\|$$

Proof: Write $V_1 = (\mathcal{G}, \rho, E')$, $V_2 = (\mathcal{G}, \rho', E')$. Then

$\|\,B_V - B_{V'}\,\| \leq [\|\,B_V - B_{V_1}\,\| + \|\,B_{V_1} - B_{V_2}\,\| + \|\,B_{V_2} - B_{V'}\,\|]$ (triangle inequality)

and $\|\,E'\,\| \leq (1+\epsilon_E)\|\,E\,\|$ but

$$\|\,B_V - B_{V_1}\,\| \leq \frac{\epsilon_E}{1-p}\|\,E\,\| \text{ (Lemma 1)}$$

$$\begin{aligned}
\|\,B_{V_1} - B_{V_2}\,\| &\leq \frac{\epsilon_\rho}{(1-p)(1-p')}\|\,E'\,\| \text{ (Lemma 3)} \\
&\leq \frac{\epsilon_\rho}{(1-p)(1-p')}(1+\epsilon_E)\|\,E\,\| \text{ (definitions)}
\end{aligned}$$

$$\|\,B_{V_2} - B_{V'}\,\| \leq \frac{(c^4-1)}{(1-p')^2}(1+\epsilon_E)\|\,E\,\| \text{ (Lemma 9)}$$

□

3 Theorem 2: Generators from Similar Scenes

We now establish that viewing shading fields for a set of similar scenes V_i is enough to estimate a representation of the generators for a given scene V. It is important to have a representation of the probability that a scene will be illuminated in some particular way. We use the general definition of a scene, above. A scene V is a tuple of geometry, albedo and luminaire model; write $V = (\mathcal{G}, \rho, \mathcal{E})$, where $\mathcal{E}$ is the luminaire model. The geometry $\mathcal{G}$ consists of a set of surfaces parametrized in some way and a 2D domain for that parametrization; write $\mathcal{G} = (\mathbf{s}(\mathbf{x}), \mathcal{D})$. The luminaire model captures the idea that lighting in a particular scene may change. We have $\mathcal{E} = (E_1, \dots E_{N_e}, P(\theta))$ for a set of potentially many basis luminaires and a probability distribution over the coefficients. This information implies a probability distribution over luminaires for the scene $P(E)$. Two scenes V and V' are similar if there is some affine transformation $\mathcal{A}, \mathbf{b}$ (smallest singular value of $\mathcal{A}$ greater than zero) such that $\mathcal{G}'(\mathcal{G}, \mathcal{A}, \mathbf{b}) = (\mathcal{A}\mathbf{s}(\mathbf{x}) + \mathbf{b})$ and if $\mathcal{E} = \mathcal{E}'$. Notice this second constraint does *not* require that the two scenes are lit in the same way, just that the basis of luminaires is the same and the probability distribution over lighting coefficients is the same.

We pass to a finite dimensional representation without loss of generality (to avoid having to deal with function spaces). Choose some N_o element orthonormal basis to represent the possible radiosity vectors for the scene (if $N_o = N_e$, the representation is exact, because there is a finite dimensional space of luminaires). In this basis, any particular radiosity $B(\mathbf{x})$ has coefficient vector $\mathbf{b}$. Write $\phi_i(\mathbf{x})$ for the i'th element of the basis; $\Phi = [\phi_1(\mathbf{x}), \dots, \phi_{N_o}(\mathbf{x})]$; $\mathbf{b}_i$ for the coefficient vector representing $B_i(\mathbf{x})$ (which is the radiosity resulting from $E_i(\mathbf{x})$); and $\mathcal{B} = [\mathbf{b}_1, \dots, \mathbf{b}_{N_e}]$.

For any $r \leq N_e$, an **effective generator matrix** for scene V is the $N_o \times r$ matrix $\mathcal{G}_{e,V,r}$ such that for any illumination condition producing radiosity represented by $\mathbf{b}$, there is some $\mathbf{w}$ such that the radiosity represented by $\mathcal{G}_{e,V,r}\mathbf{w}$ is similar to the actual radiosity field. In particular, we want

$$\mathcal{L}(\mathcal{G}_{e,V,r}) = \mathbb{E}_{\theta \sim P(\theta)}\left[\inf_{\mathbf{w}} \| \mathcal{G}_{e,V,r}\mathbf{w} - \mathcal{B}\theta \|_2^2\right]$$

to be small; $\mathcal{G}_{e,V,r}$ is clearly not unique. We choose to have $\mathcal{G}_{e,V,r}$ orthonormal.

Assume we have k radiosity fields B_i for a scene under different, unknown, illumination conditions, for k very large. We assume that for the i'th field, $\theta_i \sim P(\theta)$ (which is not known). We can clearly estimate an effective generator matrix by replacing the expectation with an average. But we do not have k images of the same scene.

Lemma 10: *Using the notation above, write*

$$C_{\mathbf{b}} = \mathcal{B}\left(\mathbb{E}_{\theta\sim P(\theta)}\left[\theta\theta^T\right]\right)\mathcal{B}^T.$$

Then:

$$\mathbb{E}_{\theta\sim P(\theta)}\left[\inf_{\mathbf{w}}\|\mathcal{G}_{e,V,r}\mathbf{w} - \mathcal{B}\theta\|_2{}^2\right]$$

is equal to

$$\mathrm{Tr}\left[C_{\mathbf{b}}\right] - \mathrm{Tr}\left[\mathcal{G}_{e,V,r}^T C_{\mathbf{b}}\mathcal{G}_{e,V,r}\right]$$

Proof: Manipulation; note that

$$C_{\mathbf{b}} = \mathbb{E}_{\theta\sim P(\theta)}\left[\mathcal{B}\theta\theta^T\mathcal{B}^T\right];$$

that

$$\mathbb{E}_{\theta\sim P(\theta)}\left[\theta^T\mathcal{B}^T\mathcal{B}\theta\right] = \mathrm{Tr}\left[C_{\mathbf{b}}\right];$$

and that

$$\mathbb{E}_{\theta\sim P(\theta)}\left[\theta^T\mathcal{B}^T\mathcal{G}_{e,V,r}\mathcal{G}_{e,V,r}^T\mathcal{B}\theta\right] = \mathrm{Tr}\left[\mathcal{G}_{e,V,r}^T C_{\mathbf{b}}\mathcal{G}_{e,V,r}\right].$$

□

We cannot *know* $\mathcal{C}_{\mathbf{b}}$, but we do not need to.

Lemma 11: *Using the notation above; writing* $\mathbf{e}_i$ *for the eigenvector of* $C_{\mathbf{b}}$ *corresponding to the* i*'th largest eigenvalue; and assuming that the eigenvalues of* $C_{\mathbf{b}}$ *are distinct, an optimal value of* $\mathcal{L}(\mathcal{G}_{e,V,r})$ *is given by*

$$[\mathbf{e}_1, \ldots, \mathbf{e}_r]\,\mathcal{M}$$

for any $r \times r$ *matrix* $\mathcal{M}$ *of full rank such that* $\mathcal{M}^T\mathcal{M} = I$

Proof: Follows directly from lemma 10.□

Lemma 12: *Using the notation above, and writing* λ_i *for the* i*'th largest eigenvalue of* $C_{\mathbf{b}}$*, the optimal value of* $\mathcal{L}(\mathcal{G}_{e,V,r})$ *is given by*

$$\sum_{i=r+1}^{N_e}\lambda_i$$

Proof: Follows directly from lemma 10 and 11. □

Assume we collect N_s samples of radiosity $\hat{\mathbf{b}} = \mathcal{B}\theta$, where $\hat{\mathbf{b}}_i$ is the i'th such sample, and $\theta \sim P(\theta)$. These yield an estimate of $C_{\mathbf{b}}$, which can be accurate.

Lemma 13: *Write* $\hat{C}_{\mathbf{b},m} = (1/N_s)\sum_i \hat{\mathbf{b}}_i\hat{\mathbf{b}}_i^T$. *Then there is some constant* C_e *such that*

$$\mathbb{E}\left[\hat{C}_{\mathbf{b},m} - C_{\mathbf{b},m}\right] = 0$$

and

$$\mathbb{E}\left[\| \hat{C}_{\mathbf{b},m} - C_{\mathbf{b},m} \|_F^2\right] = (1/N_s)C_e$$

Proof: This follows from the weak law of large numbers □

Now collect N_s samples of radiosity from different scenes V_i, each one similar to the scene V, where $\mathcal{A}_i$ maps the geometry of V to V_i. Write $\hat{\mathbf{b}}_{i,V_i}$ for the i'th such sample. Recall that similar scenes have the same basis of luminaires and the same distribution for θ. These samples yield an estimate of $C_{\mathbf{b}}$, too.

Lemma 14: *Given* $\mathbf{b}$, $\mathbf{c}$, r *and* δ *such that (a)* $\|\mathbf{b}-\mathbf{c}\|_2 \leq \delta$ *and (b)* $\|\mathbf{b}\|_2 \leq r$, *we have:*

$$\|\mathbf{b}\mathbf{b}^T - \mathbf{c}\mathbf{c}^T\|_F \leq 3\delta^2 r^2 + 4\delta^3 r + \delta^4$$

Proof: Write $\Delta = \mathbf{c} - \mathbf{b}$. Then

$$\begin{aligned}\|\mathbf{b}\mathbf{b}^T - \mathbf{c}\mathbf{c}^T\|_F &= \mathrm{Tr}\left[\left(\mathbf{b}\mathbf{b}^T - \mathbf{c}\mathbf{c}^T\right)\left(\mathbf{b}\mathbf{b}^T - \mathbf{c}\mathbf{c}^T\right)^T\right] \\ &= \begin{pmatrix} (\mathbf{b}^T\mathbf{b})^2 \\ -2(\mathbf{b}^T\mathbf{b} + \Delta^T\mathbf{b})^2 \\ +(\mathbf{b}^T\mathbf{b} + \Delta^T\mathbf{b} + \Delta^T\Delta)^2 \end{pmatrix} \\ &\leq 4\delta r^3 + 3\delta^2 r^2 + 2\delta^3 r + \delta^4\end{aligned}$$

□.

Lemma 15: *Write* $\hat{C}_{\mathbf{b},m,s} = (1/N_s)\sum_i \hat{\mathbf{b}}_{i,V_i}\hat{\mathbf{b}}_{i,V_i}^T$. *Using the notation above, and writing* $p = max_i\,(sup_{\mathcal{D}}\rho_i)$; $\epsilon_\rho = max_i\,(sup_{\mathcal{D}} abs[\rho_V - \rho_{V_i}])$; c *for the largest condition number of* $\mathcal{A}_i$. *Write*

$$\delta = \left(\frac{\epsilon_\rho + (c^4 - 1)}{(1-p)^2}\right) \text{ and } r = \frac{1}{(1-p)}$$

Then

$$\|\hat{C}_{\mathbf{b},m,s} - C_{\mathbf{b},m}\|_F \leq N_o^2\left(4\delta r^3 + 3\delta^2 r^2 + 2\delta^3 r + \delta^4\right)$$

Proof: The i'th sample θ_i from $P(\theta)$ yields a luminaire E; write $B_{i,V}$ for the resulting radiosity in V and $\mathbf{b}_{i,V}$ for its finite dimensional representation.

Theorem 1 yields a bound on $\| B_{i,V} - B_{i,V_i} \|$, and so on $\| \mathbf{b}_{i,V} - \mathbf{b}_{i,V_i} \|$. We have $\| \mathbf{b}_{i,V} - \mathbf{b}_{i,V_i} \| \leq \left(\frac{\epsilon_\rho + (c^4 - 1)}{(1-p)^2} \right)$. Furthermore, $\| \mathbf{b}_{i,V} \| \leq 1/(1-p)$. The result then follows from lemma 14 □.

We can now compute an effective generator matrix $\hat{\mathcal{G}}$ from an estimate $\mathcal{C}_{\mathbf{b},m,s}$ of $\mathcal{C}_\mathbf{b}$. This estimate uses observations of multiple similar scenes. The $\hat{\mathcal{G}}$ we obtain from data like this will not be the true $\mathcal{G}$, and we must determine what problems will result from using $\hat{\mathcal{G}}$ (which we can get) in place of $\mathcal{G}$ (which is the true best EGM).

Theorem 2 establishes that using $\hat{\mathcal{G}}$ in place of $\mathcal{G}$ incurs bounded error. Recall the expected error of using $\mathcal{G}$ as the effective generator matrix for estimate $\mathcal{C}$ is

$$\mathcal{L}(\mathcal{G}; \mathcal{C}) = \left(\text{Tr}\left[\mathcal{C}\right] - \text{Tr}\left[\mathcal{G}^T \mathcal{C} \mathcal{G}\right] \right)$$

Now estimate the effective generator matrix using $\mathcal{C}_{\mathbf{b},m,s}$ to obtain

$$\hat{\mathcal{G}} = \underset{\mathcal{G} | \mathcal{G}^T \mathcal{G} = I}{\text{argmax}} \mathcal{L}(\mathcal{G}; \mathcal{C}_{\mathbf{b},m,s})$$

Using $\hat{\mathcal{G}}$ to represent the radiosity of the scene will incur expected error

$$\mathcal{L}(\hat{\mathcal{G}}; \mathcal{C}_\mathbf{b}) = \left(\text{Tr}\left[\mathcal{C}_\mathbf{b}\right] - \text{Tr}\left[\hat{\mathcal{G}}^T \mathcal{C}_\mathbf{b} \hat{\mathcal{G}}\right] \right).$$

Theorem 2: $\mathcal{L}(\hat{\mathcal{G}}; \mathcal{C}_\mathbf{b}) - \mathcal{L}(\mathcal{G}; \mathcal{C}_\mathbf{b})$ *is bounded.*

Proof:
We have

$$\mathcal{L}(\hat{\mathcal{G}}; \mathcal{C}_\mathbf{b}) - \mathcal{L}(\mathcal{G}; \mathcal{C}_\mathbf{b}) = \text{Tr}\left[\hat{\mathcal{G}}^T \mathcal{C}_\mathbf{b} \hat{\mathcal{G}}\right] - \text{Tr}\left[\mathcal{G}^T \mathcal{C}_\mathbf{b} \mathcal{G}\right]$$

Recall $\mathcal{G}$ is chosen to minimize $\mathcal{L}(\mathcal{G}; \mathcal{C}_\mathbf{b})$, and is orthornormal, so will consist of the eigenvectors of $\mathcal{C}_\mathbf{b}$ corresponding to the r largest eigenvalues. In turn, $\hat{\mathcal{G}}$ is chosen to minimize $\mathcal{L}(\hat{\mathcal{G}}; \mathcal{C}_{\mathbf{b},m,s})$. We must now find the $\hat{\mathcal{G}}$ so that $\text{Tr}\left[\mathcal{G}^T \mathcal{C}_\mathbf{b} \mathcal{G}\right]$ is minimized within the bound on $\mathcal{C}_\mathbf{b} - \mathcal{C}_{\mathbf{b},m,s}$. Any extremal $\hat{\mathcal{G}}$ must consist of eigenvectors of $\mathcal{C}_\mathbf{b}$ (recall $\hat{\mathcal{G}}$ is orthornormal). This yields an immediate but rather loose bound. Label the eigenvalues λ_i of $\mathcal{C}_\mathbf{b}$ so that $\lambda_1 \geq \lambda_2 \geq \ldots \lambda_{N_o}$. then

$$\inf_{\hat{\mathcal{G}} | \hat{\mathcal{G}}^T \hat{\mathcal{G}} = \mathcal{I}} \text{Tr}\left[\mathcal{G}^T \mathcal{C}_\mathbf{b} \mathcal{G}\right] = \sum_{i=N_o - r + 1}^{N_o} \lambda_i$$

so

$$\mathcal{L}(\hat{\mathcal{G}}; \mathcal{C}_\mathbf{b}) - \mathcal{L}(\mathcal{G}; \mathcal{C}_\mathbf{b}) \leq \sum_{i=1}^{N_o} \lambda_i - \sum_{i=N_o - r + 1}^{N_o} \lambda_i.$$

□

This bound is quite loose, but for the purposes of this paper, the existence of a bound is enough. Some quite interesting mathematics is possible in tightening the bound, and there may be practical consequences.

Notice that, writing $\Phi(\mathcal{G};\mathcal{C}) = \text{Tr}\left[\mathcal{C}\right] - \text{Tr}\left[\mathcal{G}^T\mathcal{C}\mathcal{G}\right]$, with $\mathcal{C}$ symmetric, diagonalized as $\mathcal{U}^T\Lambda\mathcal{U}$; then $\Phi(\mathcal{U}\mathcal{G};\Lambda) = \Phi(\mathcal{G};\mathcal{C})$. This means we can work with diagonal $\mathcal{C}$ WLOG. In turn, determining the value of $\mathcal{L}(\hat{\mathcal{G}};\mathcal{C}_\mathbf{b}) - \mathcal{L}(\mathcal{G};\mathcal{C}_\mathbf{b})$ requires thinking about shuffling eigenvalues. Any bound will follow from reasoning about permutations in the order of eigenvectors achieved by the error, and the estimation error may not be large enough to permute the eigenvalues so aggressively. A more interesting and tighter bound can be obtained as

$$\sum_{i=1}^{N_o} \lambda_i - v$$

where v is the value of a linear program representing the permuation of the eigenvalues. Write $\mathbf{1}_r$ for the N_o dimensional vector consisting of r ones and $N_o - r$ zeros in order, $\lambda = [\lambda_1, \ldots, \lambda_{N_o}]$ (eigenvalues of $\mathcal{C}_\mathbf{b}$), $\mathcal{P}$ for a permutation matrix, δ for a vector of offsets, D for $\|\mathcal{C}_\mathbf{b} - \mathcal{C}_{\mathbf{b},m,s}\|_F{}^2$, then v is the value of:

$$\begin{aligned}
&\min_{\mathcal{P},\delta,\mathbf{m}} \mathbf{1}_r^T\mathcal{P}\lambda \\
&\text{subject to} \\
&\quad \sum_i p_{ij} = 1 \\
&\quad \sum_j p_{ij} = 1 \\
&\quad p_{ij} \geq 0 \\
&\quad \mathbf{u} = \mathcal{P}\lambda \\
&\quad \mathbf{m} = \mathbf{u} + \delta \\
&\quad m_1 \geq m_2 \geq \ldots m_{N_o} \\
&\quad \sum_i \delta_i^2 \leq D.
\end{aligned}$$

This problem finds the permutation of eigenvalues that produces the smallest value within the budget D. For $D \geq 0$, the program is feasible ($\mathcal{P} = \mathcal{I}$ is always a feasible point), and Birkhoff's theorem guarantees that a there is a solution with integer $\mathcal{P}$. Interestingly, for sufficiently small D, $v = \sum_{i=1}^{N_o} \lambda_i$, meaning the error bound is zero! This occurs when D is smaller than $\lambda_{N_o} - \lambda_{N_o+1}$. The quantized character of this bound means that quite small improvements in estimation of $\mathcal{C}_{\mathbf{b},m,s}$ could result in large improvements in the practical behavior of the method.

4 The Local FID

Definition: Assume we obtain a collection of images $\mathcal{R}$ from a ground truth collection $\mathcal{O}$ by relighting a single image I_k and compute FID$(\mathcal{O},\mathcal{R})$. This value

– which is non-negative, and is ideally zero – measures the extent to which the relighted image is realistic. Furthermore, different relightings of the same image will have different values. Write μ_O for the mean and C_O for the covariance of $\mathcal{O}$ in the chosen embedding, etc. We have

$$\begin{aligned}\text{FID}(\mathcal{O}, \mathcal{R}) = &\| \mu_R - \mu_O \|_2^2 + \\ &\text{Trace}(C_O + C_R - 2(C_O C_R)^{(1/2)}).\end{aligned}$$

Computing difficulties: Numerical problems occur because C_R is very similar to C_O, and C_O has many small eigenvalues; this means there are many eigenvalues of $C_O C_R$ that are very close to zero, which is a reliable source of trouble for matrix square root methods. One typically gets complex estimates.

Series estimate: Write $\mathbf{x}(I_k)$ for the embedding of I_k, R_k for the result of relighting I_k, and $\mathbf{d}_k = \mathbf{x}(R_k) - \mathbf{x}(I_k)$. Then $\mu_R - \mu_O = \frac{\mathbf{d}_k}{N}$, and

$$\begin{aligned}C_R &= C_O \\ &\quad + \frac{1}{N}\begin{pmatrix}\mathbf{d}_k(\mathbf{x}(I_k) - \mu_0)^T + \\ (\mathbf{x}(I_k) - \mu_0)\mathbf{d}_k^T + \\ \mathbf{d}_k\mathbf{d}_k^T\end{pmatrix} \\ &\quad - \frac{1}{N^2}\mathbf{d}_k\mathbf{d}_k^T \\ &= C_O + \frac{1}{N}U + \frac{1}{N^2}V\end{aligned}$$

We seek an approximation to $(C_O C_R)^{(1/2)}$ of the form $C_O + \frac{1}{N}M_1 + \frac{1}{N^2}M_2 + \ldots$. By squaring and matching terms, we have

$$C_O M_1 + M_1 C_O = C_O U$$

and

$$C_O M_2 + M_2 C_O + M_1 M_1 = C_O V$$

so $2\text{Trace}(M_1) = \text{Trace}(U)$ and first order terms cancel. Similarly, $\text{Trace}(V) = 2\text{Trace}(M_2) + \text{Trace}(C_O^{-1} M_1^2)$, and we have

$$\text{FID}(\mathcal{O}, \mathcal{R}) = \frac{1}{N^2}(\mathbf{d}_k^T \mathbf{d}_k + \text{Trace}(C_O^{-1} M_1^2) + O(\frac{1}{N}^3)$$

and we define the local FID at the k'th image as

$$\text{FID}_{l,\mathcal{O}}(I_k) = (\mathbf{d}_k^T \mathbf{d}_k + \text{Trace}(C_O^{-1} M_1^2)).$$

But we must evaluate M_1. We have $C_O M_1 + M_1 C_O = C_O U$, which is a symmetric Sylvester equation. While this is a linear system in M_1, it is inconveniently large. But, for R a rotation and G general, $(RGR^T)^{(1/2)} = R(G)^{(1/2)}R^T$, so

$$\text{Trace}(RC_O R^T + RC_R R^T - 2(RC_O R^T RC_R R^T)^{(1/2)})$$

is equal to

$$\text{Trace}(C_O + C_R - 2(C_O C_R)^{(1/2)})$$

and the FID is not affected by a rotation in the embedding space, so we can assume without loss of generality that C_O is diagonal. In this case, we can compute M_1; write a_{ij} for the i, j'th component of A etc., and we have $m_{ij} = c_{ii}u_{ij}/(c_{ii} + c_{jj})$, and computing the local FID at a particular image is relatively straightforward.

LFID is a good predictor of human preference: We show human subjects a pair of images (A, B) *of the same scene*, and record which is regarded as more realistic. Images are chosen uniformly at random from the original, and four relightings. We have 5000 runs of 10 choices, with anonymous subjects. The scatter plot shows frequency that A is preferred against LFID(A)-LFID(B) for 500 quantiles of the difference (so frequencies are estimated with 100 points). Subjects reliably choose the image with lower LFID, though probabilities at large differences saturate. The variation in frequencies suggest that other predictors might be required to score realism of an image transformation more accurately.

5 The Gloss Term

Experimental results suggest strongly our method can remove or enhance glossy effects. Theorem 1 covers global illumination in a diffuse world. The methods used to prove this theorem extend to cover single-bounce radiance from a luminaire for some BRDF models, including natural gloss models, if all BRDF's have the property

$$\left[(\omega' \cdot \omega)(\overline{\omega'} \cdot \overline{\omega})\right]^{\alpha} \leq \frac{b(\mathbf{x}, \omega', \overline{\omega}')}{b(\mathbf{x}, \omega, \overline{\omega})} \leq \left[\frac{1}{(\omega' \cdot \omega)(\overline{\omega'} \cdot \overline{\omega})}\right]^{\alpha}.$$

Notice all diffuse surfaces have this property for any finite $\alpha \geq 0$.